

Customer Segmentation Analysis

1. Project Overview

A leading retail company wants to better understand its customers by identifying distinct groups based on their shopping behavior to improve sales, customer satisfaction, and long-term loyalty. The management team has noticed changes in purchasing patterns across demographics, product categories, and sales channels (online s. offline), and they want to use customer segmentation to uncover these patterns. These insights will enable the business to develop targeted marketing campaigns, personalise promotions, and improve customer retention.

2. Problem Statement

Retail businesses often apply the same marketing strategies to all customers, regardless of the differences in preferences and purchase behavior. This project aims to use unsupervised machine learning techniques to segment customers into meaningful groups. The resulting customer segments will help the business better understand its customer base and support data-driven decisions to optimize marketing and product strategies.

3. Tools Used

- Python
- SQL
- PowerBI
- GitHub
- Gammas
- Comet

4. Dataset Overview

- Rows: 3,900
- Columns: 18
- Key Features:
 - Customer demographics (Age, Gender, Location, Subscription Status)
 - Purchase details (Item Purchased, Category, Purchase Amount, Season, Size, Color)
 - Shopping behavior (Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type)
- Missing Data: 37 values in the Review Rating column

5. Data Cleaning

Start by preparing and cleaning the data in Python.

→ **Data Loading:** Imported the data using [pandas](#).

→ **Initial Data Exploration:** Used [df.info\(\)](#) to check the dataset's structure and [describe\(\)](#) to get summary statistics.

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Promo Code Used	Previous Purchases	Payment Method	Frequency of Purchases
count	3900.000000	3900.000000	3900	3900	3900	3900.000000	3900	3900	3900	3900	3863.000000	3900	3900	3900	3900	3900.000000	3900	3900
unique	NaN	NaN	2	25	4	NaN	50	4	25	4	NaN	2	6	2	2	NaN	6	7
top	NaN	NaN	Male	Blouse	Clothing	NaN	Montana	M	Olive	Spring	NaN	No	Free Shipping	No	No	NaN	PayPal	Every 3 Months
freq	NaN	NaN	2652	171	1737	NaN	96	1755	177	999	NaN	2847	675	2223	2223	NaN	677	584
mean	1950.500000	44.068462	NaN	NaN	NaN	59.764359	NaN	NaN	NaN	NaN	3.750065	NaN	NaN	NaN	NaN	25.351538	NaN	NaN
std	1125.977353	15.207589	NaN	NaN	NaN	23.685392	NaN	NaN	NaN	NaN	0.716983	NaN	NaN	NaN	NaN	14.447125	NaN	NaN
min	1.000000	18.000000	NaN	NaN	NaN	20.000000	NaN	NaN	NaN	NaN	2.500000	NaN	NaN	NaN	NaN	1.000000	NaN	NaN
25%	975.750000	31.000000	NaN	NaN	NaN	39.000000	NaN	NaN	NaN	NaN	3.100000	NaN	NaN	NaN	NaN	13.000000	NaN	NaN
50%	1950.500000	44.000000	NaN	NaN	NaN	60.000000	NaN	NaN	NaN	NaN	3.800000	NaN	NaN	NaN	NaN	25.000000	NaN	NaN
75%	2925.250000	57.000000	NaN	NaN	NaN	81.000000	NaN	NaN	NaN	NaN	4.400000	NaN	NaN	NaN	NaN	38.000000	NaN	NaN
max	3900.000000	70.000000	NaN	NaN	NaN	100.000000	NaN	NaN	NaN	NaN	5.000000	NaN	NaN	NaN	NaN	50.000000	NaN	NaN

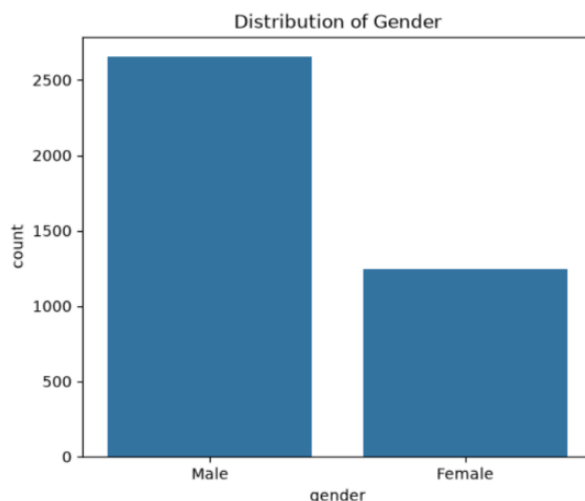
→ **Handling Missing Values:** Imputed missing values for [Review_Rating](#) using the category-specific median.

→ **Column Standardization:** Renamed all columns to **snake case** for better documentation.

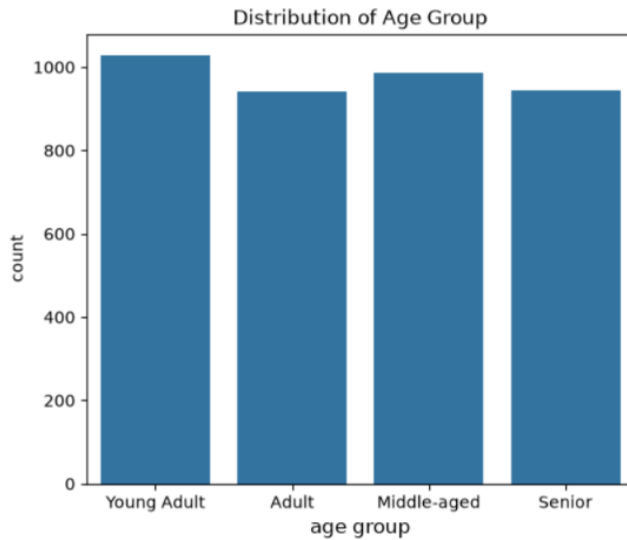
→ **Data Consistency Check:** Verified if [promo_code_used](#) and [discount_applied](#) were redundant. Dropped the [promo_code_used](#) column since both columns carry the same information.

6. Exploratory Data Analysis

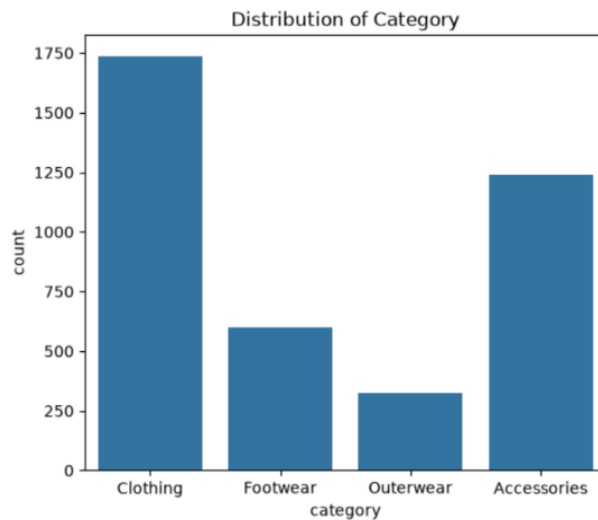
Univariate Analysis



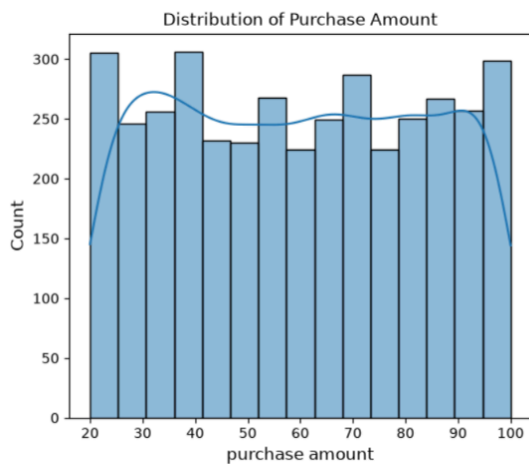
The dataset is heavily skewed toward males with a 2:1 ratio, meaning the number of males is more than double that of females.



The age group feature shows a highly balanced and uniform distribution, with the young_adult category peaking just above 1000.

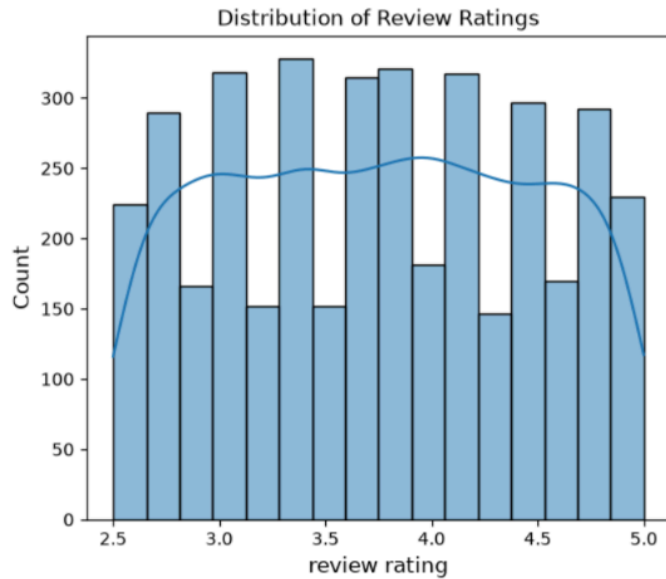


The category that sells the most by far is clothing. The core inventory or customer purchase volume is driven by clothing and accessories.

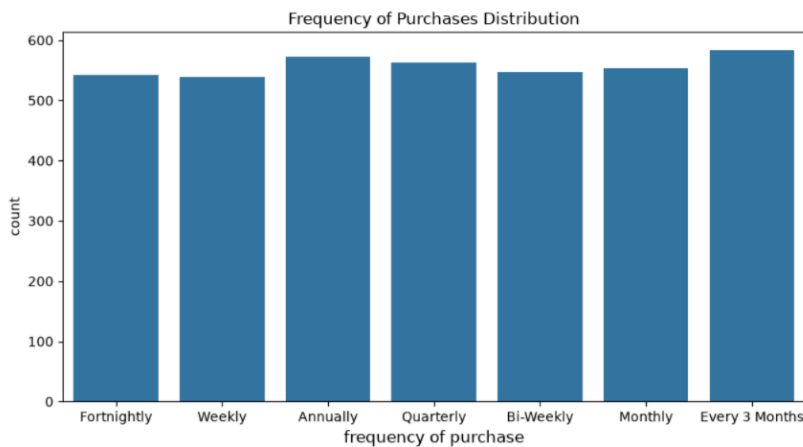


purchase_amount	
count	3900
mean	59.7644
std	23.6854
min	20
25%	39
50%	60
75%	81
max	100

Spending is not normally distributed. Purchase amounts are strictly bounded between 20 and 100. On average, a customer spends roughly \$60 per purchase.

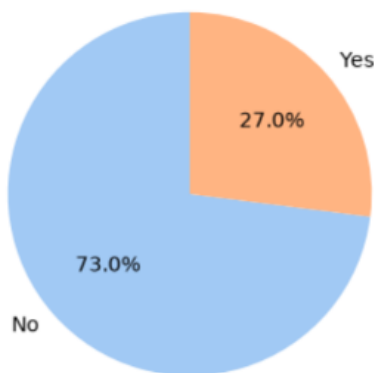


The ratings range from 2.5 to 5, with observations spread fairly evenly. There are no obvious peaks or signs of strong skewness. This suggests that customer satisfaction varies among purchases, making ratings a potentially useful feature for identifying segments. The KDE shows a slight concentration of ratings around 3.8 to 4, indicating that moderately positive reviews are somewhat more common than the lower ones.



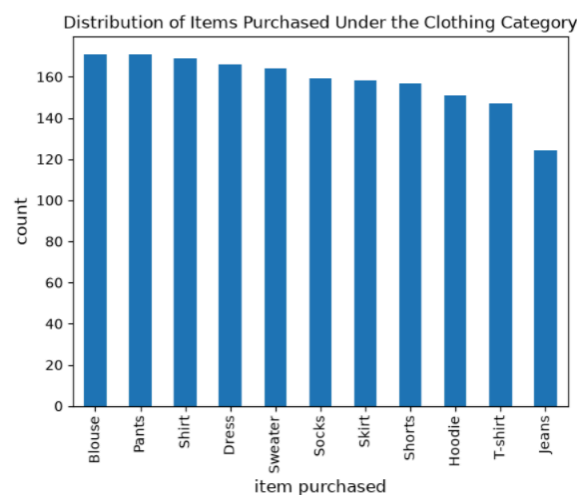
The distribution is fairly even across all categories; the differences between categories are minimal.

Subscription Distribution

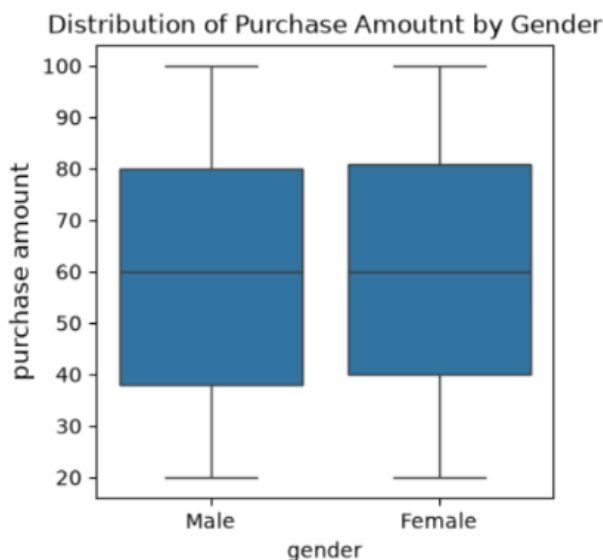
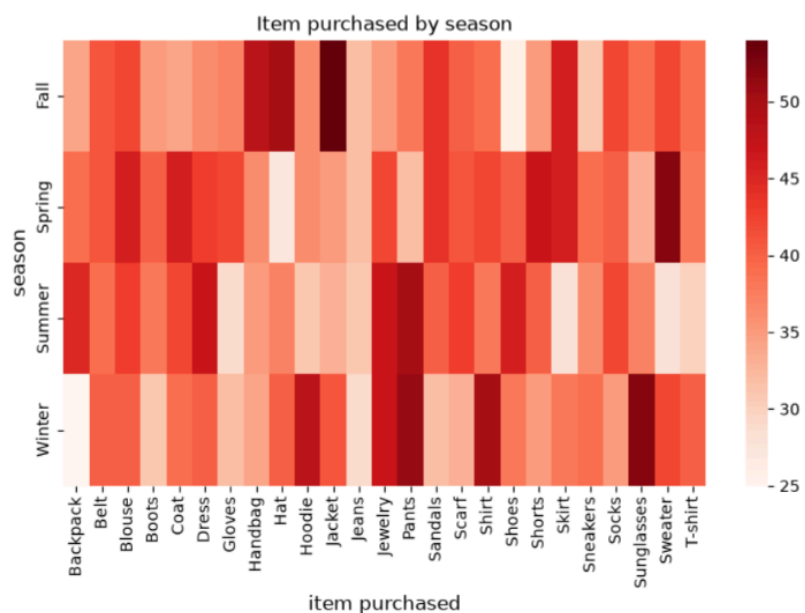


73% of the customers are subscribed, indicating a high customer adoption rate. 27% of customers are not subscribed, indicating a potential market for future subscription growth.

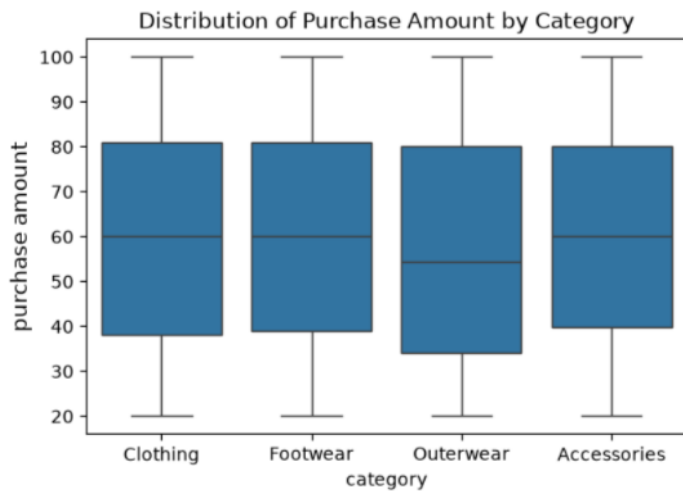
Bivariate Analysis



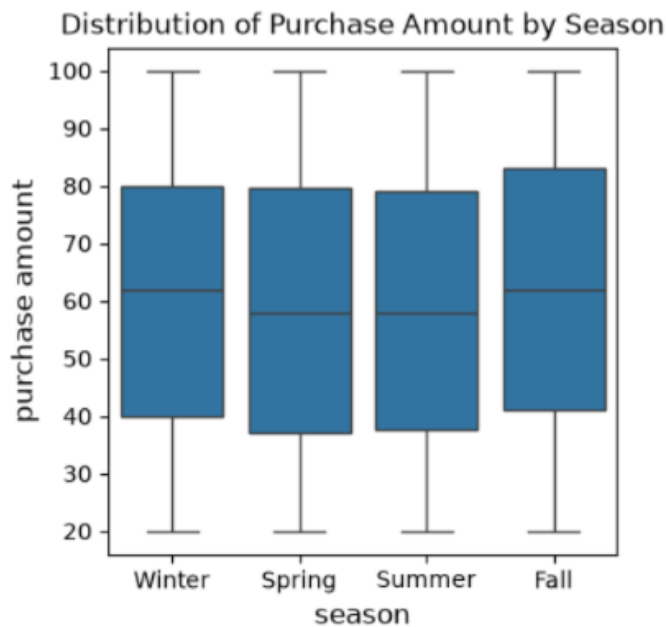
As indicated in the previous graph, clothing is the top-selling category, with blouses, pants, and shirts emerging as the three most popular items.



Although the data has more male customers than female customers, there are no major differences in median purchase amount between the two groups.



Clothing, footwear, and accessories have the same median purchase amounts. So customers usually spend around the same amount when they purchase items under these categories. Outerwear has the lowest median purchase amount, indicating that customers generally spend less on outerwear purchases

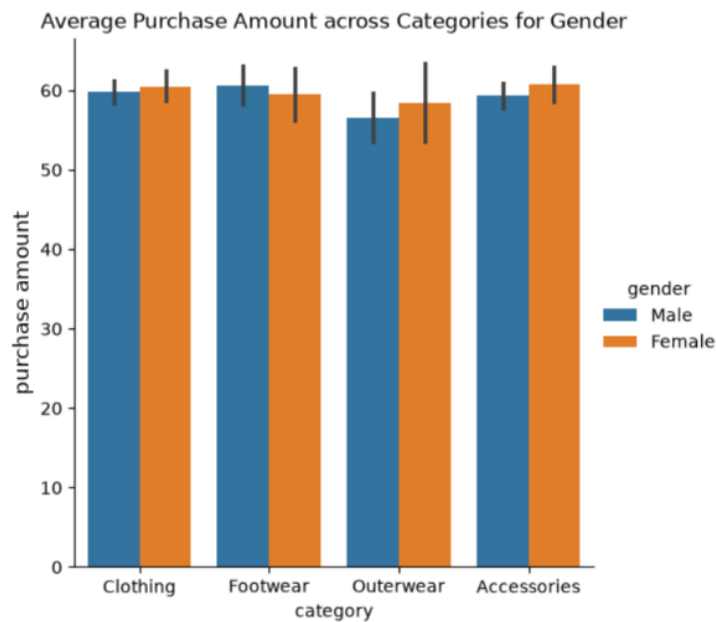


Seasonal spending patterns show that customers spend more in winter and fall, whereas spending decreases during the warmer months.

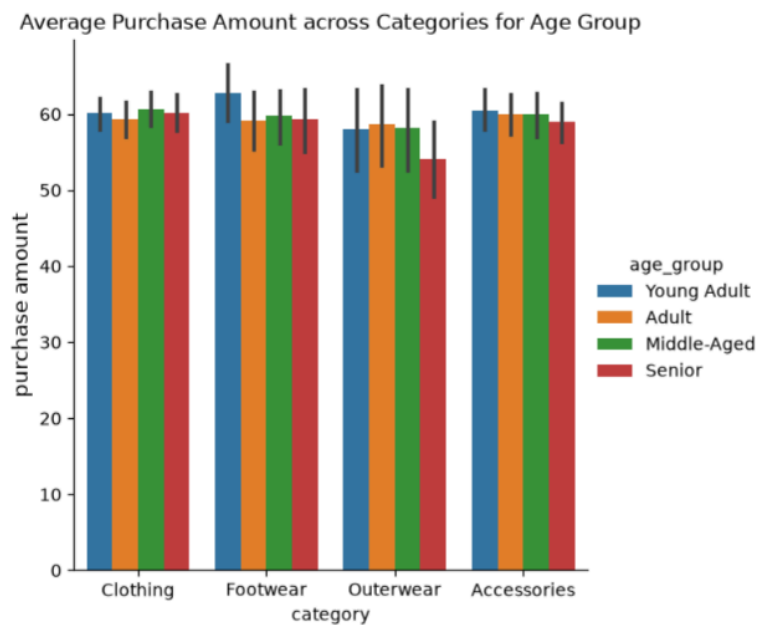
Purchase Amount vs other features.

Customers generally spend the same amount among these features: [subscription_status](#), [gender](#), and [discount_applied](#).

Multivariate Analysis

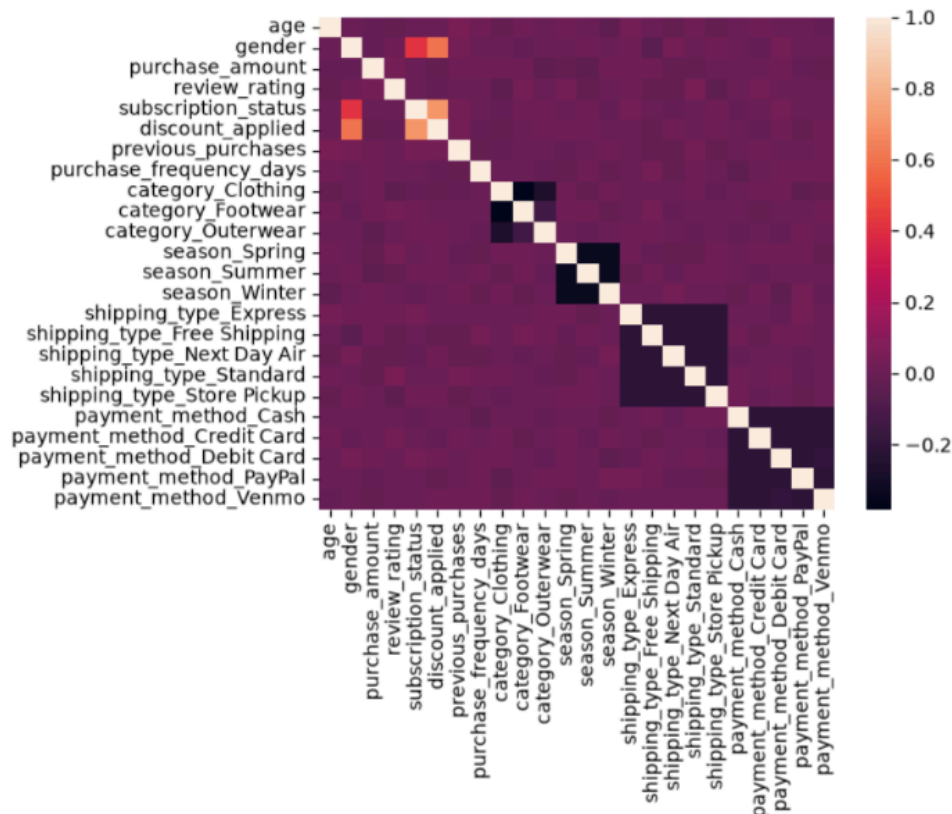


Spending patterns are generally similar between genders across all categories. Female customers tend to spend slightly more on clothing, outerwear, and accessories, while male customers spend more on footwear. The overall differences are small, suggesting that gender might have limited influence on average spending within each category.



All age groups spend similar amounts on clothing. Young adults have the highest average purchase amount for footwear. Outerwear has the lowest average purchase amount across all age groups. Senior customers spend noticeably less on outerwear than the other age groups. Young adults and middle-aged customers spend marginally more than adults and seniors.

Correlation Analysis Using a Heatmap



This correlation heatmap shows the relationships between the encoded variables in the dataset. Most variables show a weak correlation with values close to zero. This indicates that the features are largely independent and there is no serious multicollinearity, which is desirable for modeling. Customers with subscriptions appear more likely to receive or use discounts. Customer demographics and purchasing behavior do not exhibit strong linear relationships with the encoded categorical variables.

Summary of EDA

The exploratory analysis revealed that customer spending varies across product categories and subscription status. Customers exhibit different purchase frequencies and previous purchase histories, suggesting potential behavioral differences suitable for clustering. Purchase amount and purchase frequency provide strong indicators of customer engagement. These findings support the use of clustering techniques to identify distinct customer segments for targeted marketing strategies.

7. Key Findings using SQL

SSMS was used to answer key business questions:

1. **Revenue by Gender** – Compared total revenue generated by male vs. female customers.

	gender	revenue
1	Male	157890
2	Female	75191

2. **High-Spending Discount Users** – Identified customers who used discounts but still spent above the average purchase amount.

	customer_id	purchase_amount
1	2	64
2	3	73
3	4	90
4	7	85
5	9	97
6	12	68
7	13	72
8	16	81
9	20	90
10	22	62
11	24	88
12	29	94

Row: 1, Col: 1 | 863 rows

3. **Top 5 Products by Rating** – Found products with the highest average review ratings.

	item_purchased	Average Product Rating
1	Gloves	3,86
2	Sandals	3,84
3	Boots	3,82
4	Hat	3,8
5	Skirt	3,78

4. **Shipping Type Comparison** – Compared average purchase amounts between Standard and Express shipping.

	shipping_type	Average Purchase Amount
1	Express	60
2	Standard	58

5. **Subscribers vs. Non-Subscribers** – Compared average spend and total revenue across subscription status.

	subscription_status	total_customers	avg_spend	total_revenue
1	Yes	1053	59	62645
2	No	2847	59	170436

6. **Discount-Dependent Products** – Identified 5 products with the highest percentage of discounted purchases.

	item_purchased	discount_rate
1	Hat	50
2	Coat	49
3	Sneakers	49
4	Sweater	48
5	Pants	47

7. **Customer Segmentation** – Classified customers into New, Returning, and Loyal segments based on purchase history.

	customer_segment	Number of Customers
1	New	83
2	Returning	701
3	Loyal	3116

8. **Top 3 Products per Category** – Listed the most purchased products within each category.

	item_rank	category	item_purchased	total_orders
1	1	Accessories	Jewelry	171
2	2	Accessories	Belt	161
3	3	Accessories	Sunglasses	161
4	1	Clothing	Blouse	171
5	2	Clothing	Pants	171
6	3	Clothing	Shirt	169
7	1	Footwear	Sandals	160
8	2	Footwear	Shoes	150
9	3	Footwear	Sneakers	145
10	1	Outerwear	Jacket	163
11	2	Outerwear	Coat	161

9. **Repeat Buyers & Subscriptions** – Checked whether customers with >5 purchases are more likely to subscribe.

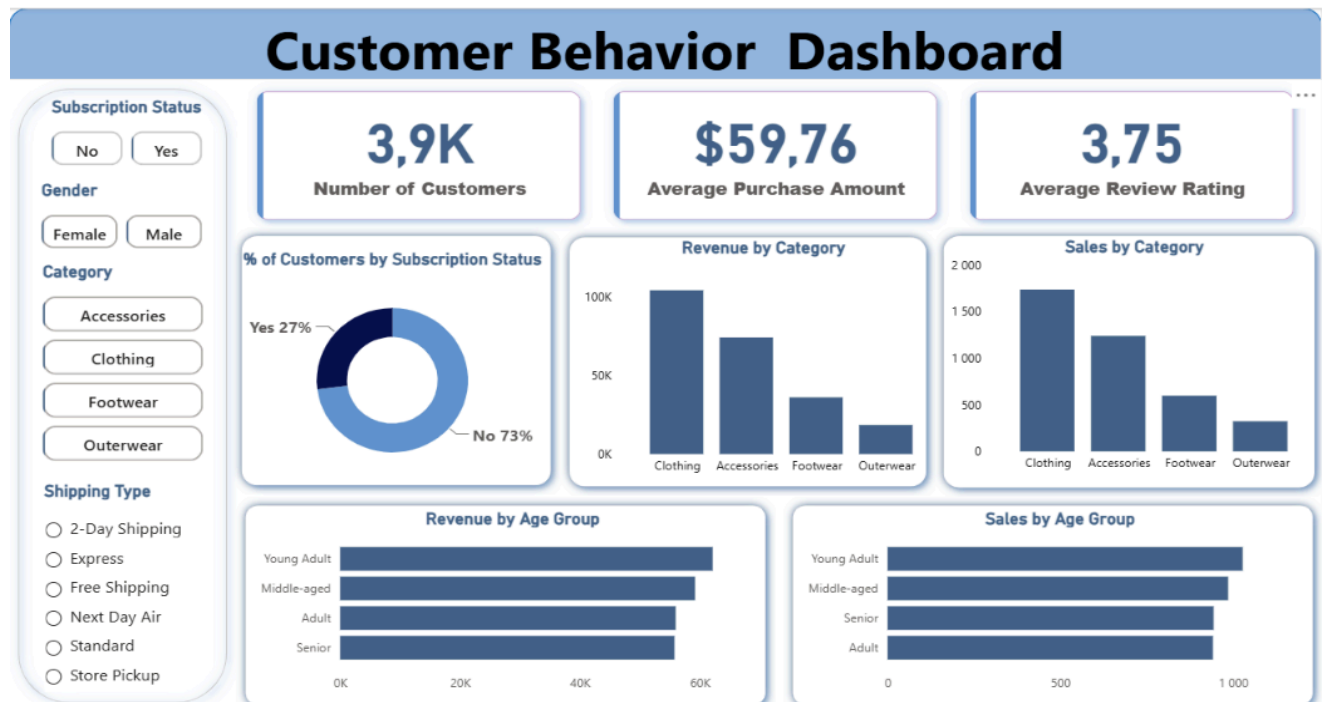
	subscription_status	repeat_buyers
1	Yes	958
2	No	2518

10. **Revenue by Age Group** – Calculated total revenue contribution of each age group.

	age_group	total_revenue
1	Young Adult	62143
2	Middle-Aged	59197
3	Adult	55978
4	Senior	55763

8. Dashboard in Power BI

Interactive dashboard built with Power BI.



9. Feature Engineering

Features that will be used for modeling

Ready for use:

- * **age** - Important demographic feature.
- * **purchase_amount** - Strong segmentation variable.
- * **previous_purchases** - Excellent measure of customer loyalty.
- * **review_rating** - Represents customer satisfaction.
- * **purchase_frequency_days** - Informative behavioral feature.

Encoded:

- * **gender** - Can help distinguish customer groups.
- * **category** - Reflects product preference.

	purchase_frequency_days	frequency_of_purchases
0	14	Fortnightly
1	14	Fortnightly
2	7	Weekly
3	7	Weekly
4	365	Annually
5	7	Weekly
6	90	Quarterly

1. Created a new feature, `purchase_frequency_days`, by frequency mapping the number of days for the `frequency_purchases` labels.

2. Created a new feature, `high_spender`. Customers who spend more than the average purchase amount are classified as high spenders.

Scaling

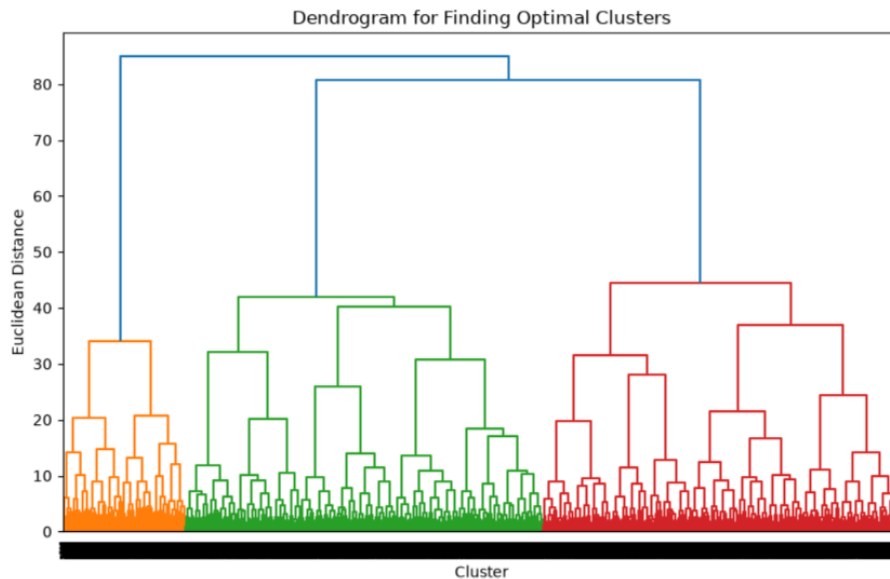
Scaling prevents large-scale features from dominating smaller ones, helps models train faster, and improves overall prediction accuracy. To scale the data, z-score scaling was used.

Z-score Standardization: The values in the feature column are standardized by removing the mean and scaling the variance to 1. This method is known as Scaling or Centering.

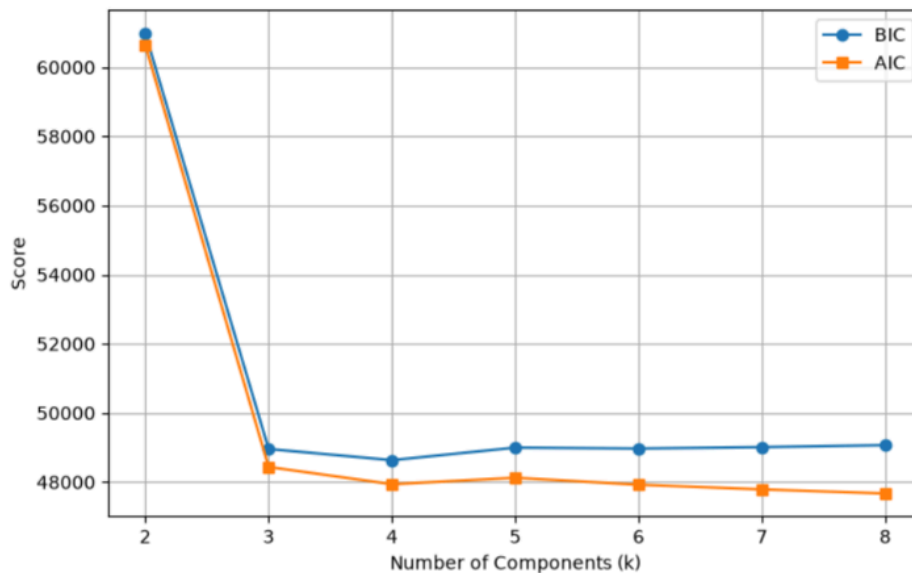
Standardized `age`, `purchase_amount`, `review_rating`, `previous_purchases`, and `purchase_frequency_days`

	age	purchase_amount	review_rating	previous_purchases	purchase_frequency_days
0	0.718913	-0.285629	-0.911076	-0.785831	-0.631252
1	-1.648629	0.178852	-0.911076	-1.616552	-0.631252
2	0.390088	0.558882	-0.911076	-0.162789	-0.690065
3	-1.517099	1.276716	-0.350458	1.637107	-0.690065
4	0.061263	-0.454531	-1.471695	0.391025	2.317775

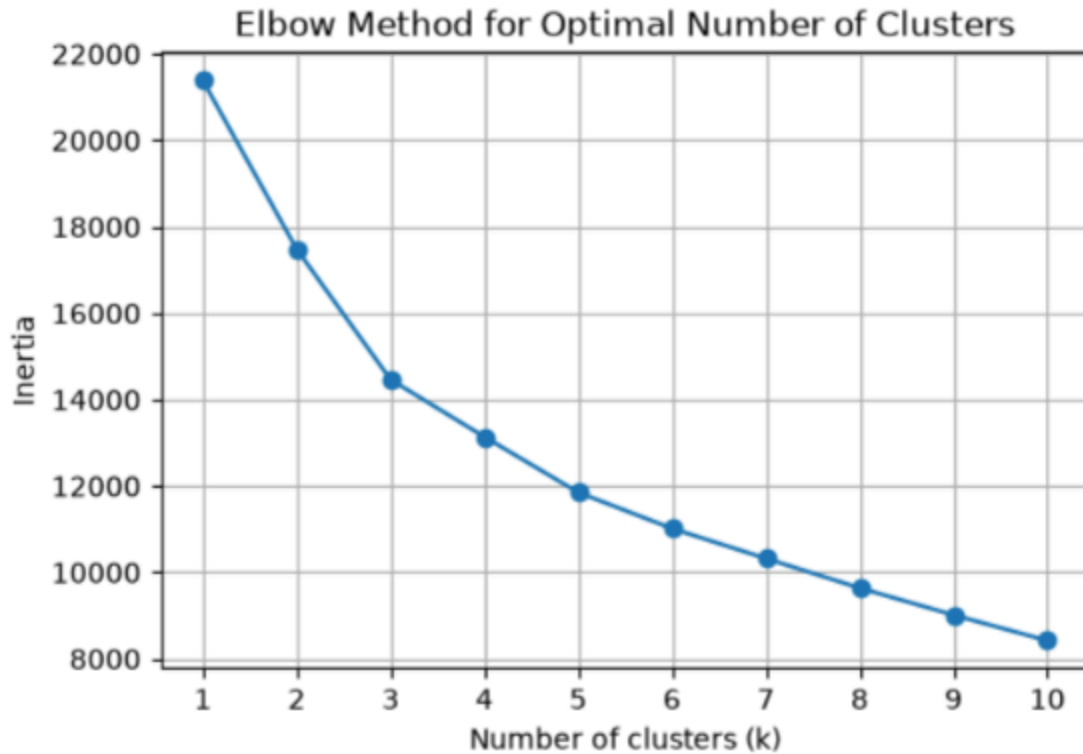
10. Determine the Value of k



The dendrogram reveals three distinctly colored primary branches (orange, green, and red), indicating a clear separation of the data. Thus, a three-cluster solution was determined to be optimal for this dataset. This choice effectively balances cluster homogeneity with model simplicity.



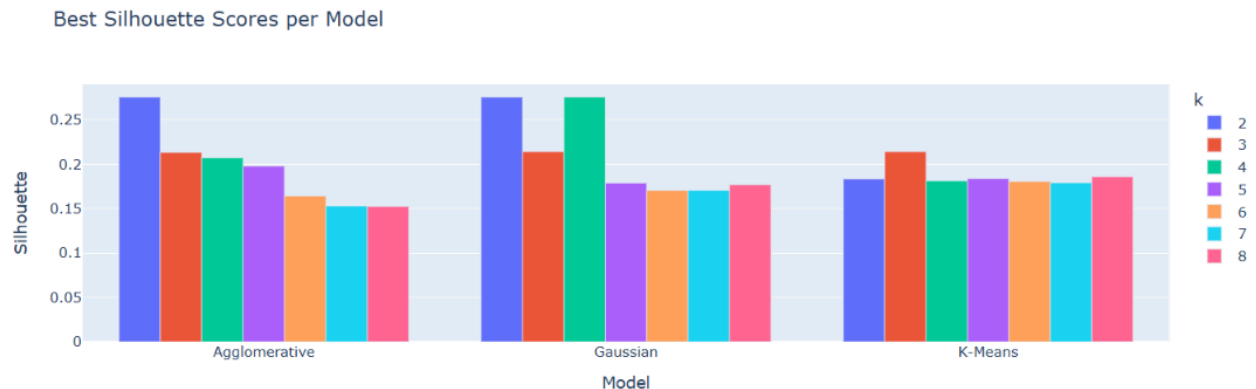
When AIC and BIC point to different numbers, BIC is generally preferred for cluster selection because it is asymptotically consistent and prevents over-clustering. Since BIC clearly minimizes at $k = 4$ and AIC shows a major elbow/inflection point there, $k = 4$ is the optimal number of components.

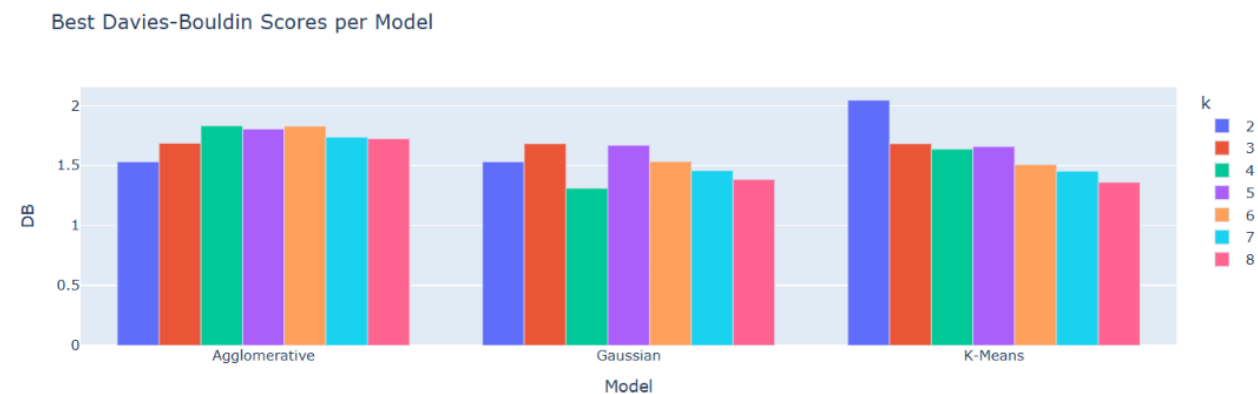
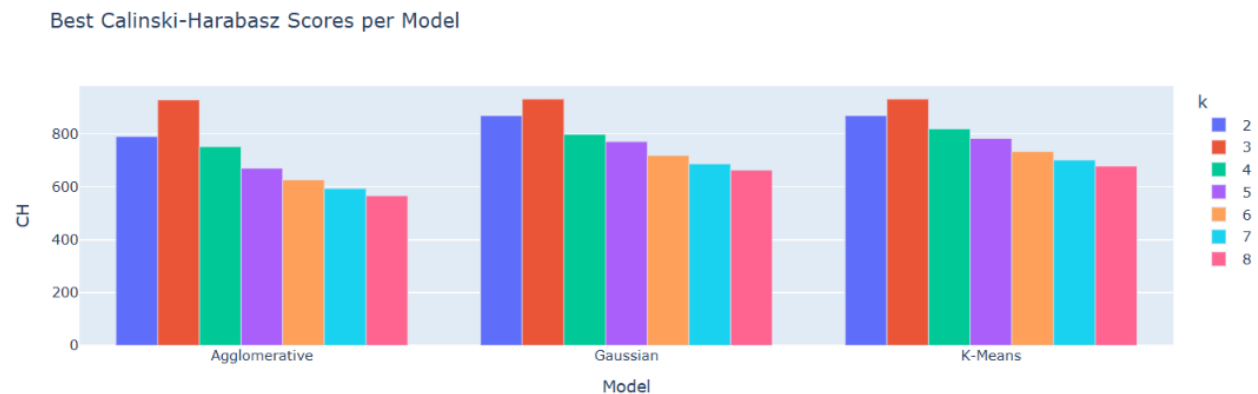


A three-cluster or four-cluster segmentation would be optimal.

11. Modeling

In this project, demographic and behavioral segmentation were used to provide the company with insights to determine the most effective way to spend its marketing budget. Agglomerative Hierarchical Clustering, Gaussian Mixture, and K-Means models were used to segment the customers.





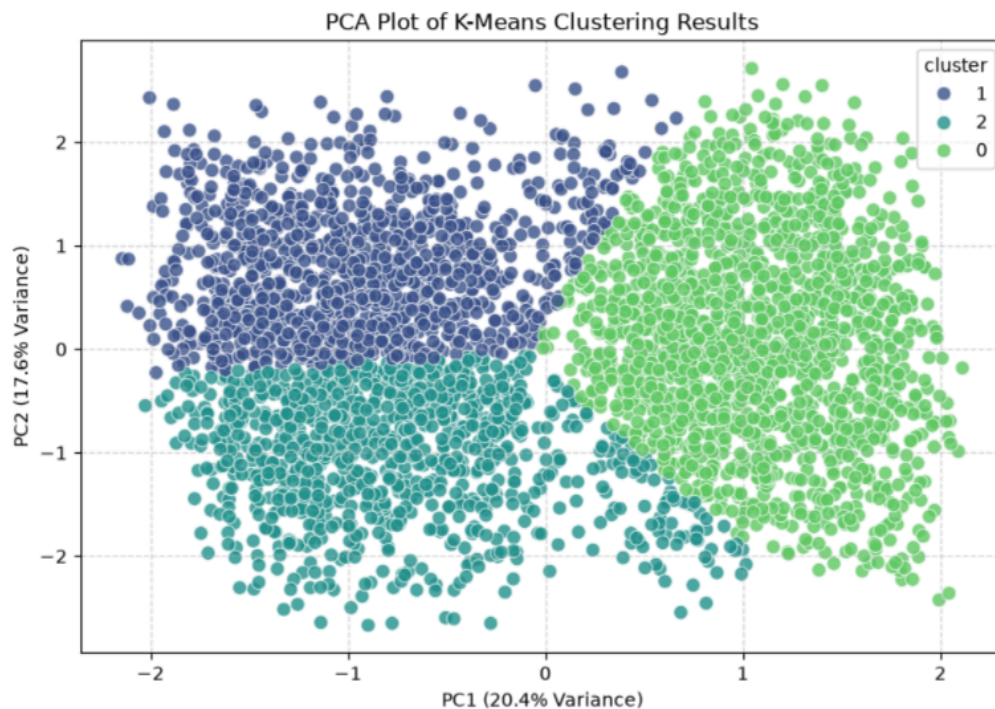
Three clustering algorithms, K-Means, Agglomerative Clustering, and Gaussian Mixture Models (GMM), were evaluated using the Silhouette Score, Calinski-Harabasz Index, and Davies-Bouldin Index. Although both Agglomerative Clustering and Gaussian Mixture Models performed well across the indices, K-Means consistently produced the lowest Davies-Bouldin Index, and it demonstrated the most balanced overall performance, indicating more compact and well-separated clusters. The performance differences between the algorithms were relatively small, suggesting that all three produced meaningful cluster structures.

Considering its competitive performance across the evaluation metrics, computational efficiency, ease of interpretation and profiling, and widespread use in customer segmentation, K-Means was selected as the final clustering algorithm, with k=3.

Final model to be used to segment the customers:

Variance	Model	k	Linkage	Silhouette	CH	DB	init	n_init	max_iter
0.9	K-Means	3	NaN	0.214087	932.184255	1.681664	random	50.0	500.0

12. Cluster Profiling



The plot shows that the clusters are reasonably well separated, indicating that the K-Means algorithm successfully identified distinct customer segments.

cluster	count	age	purchase_amount	review_rating	previous_purchases	purchase_frequency_days	total_revenue
0	1635	43.907645	80.607951	3.773333	25.720489	42.221407	131794
1	1693	44.021855	39.496751	3.725103	25.262256	41.232723	66868
2	572	44.666084	60.173077	3.757343	24.561189	365.000000	34419

Cluster 0 - High-Value Loyal Customers

This cluster represents the business's most valuable customers. They made the highest number of previous purchases, indicating strong customer loyalty and a high likelihood of repeat business. They also spend relatively large amounts per purchase while providing the highest review ratings, suggesting a high level of satisfaction with the products they purchase. Additionally, they purchase more than the other clusters. This segment is essential for maintaining revenue and

Cluster 1 - Budget-Conscious Regular Customers

Cluster 1 consists of customers who spend the least amount per transaction despite having a relatively high number of previous purchases. Their review ratings indicate a generally positive experience, providing an opportunity to strengthen the relationship with this segment. They appear to be price-sensitive and may respond well to discounts, coupons, bundled offers, or promotional campaigns aimed at increasing purchase value and improving customer satisfaction. They are also a good target for upselling.

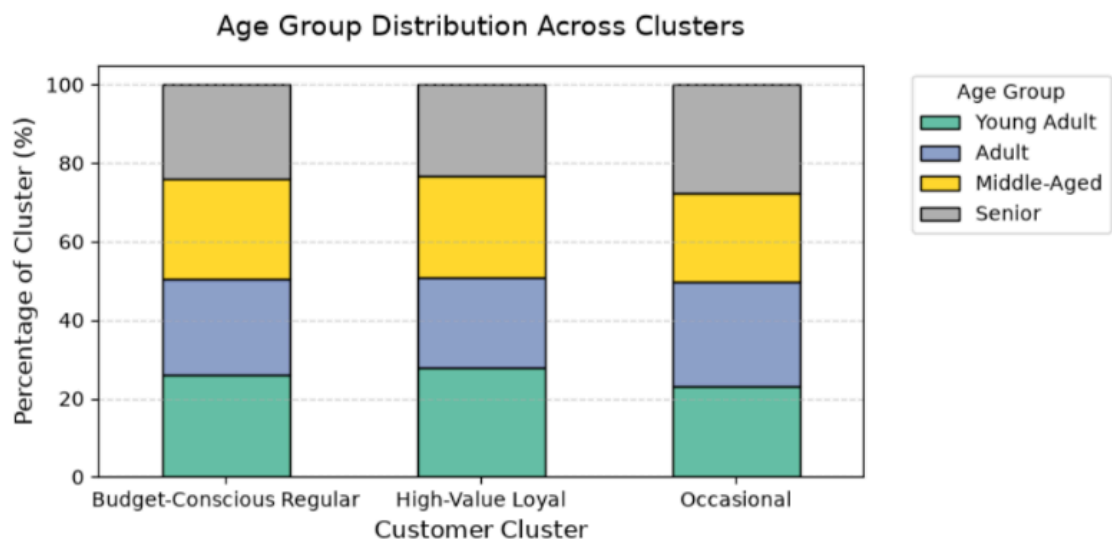
Cluster 2 - Occasional Customers

This cluster includes customers who spend relatively high amounts when they make a purchase but have made relatively few purchases overall. They have the lowest overall revenue because of the small size and infrequent purchases. Their review ratings indicate a generally positive experience, providing an opportunity to strengthen the relationship with this segment. These customers have the potential to become loyal, high-value customers if encouraged through personalized recommendations, targeted promotions, and retention campaigns.



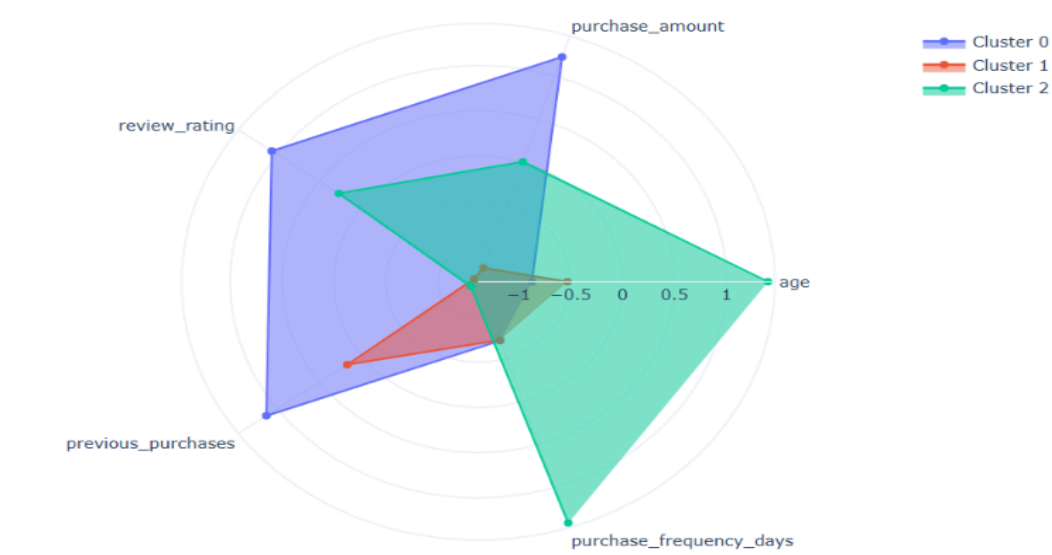
The model successfully separated the absolute least frequent buyers into their own distinct segment. They only buy once a year. Purchase frequency does not differentiate these two segments. Both groups are active, highly regular shoppers who buy at various intervals throughout the year. Because the Budget-Conscious and High-Value segments look identical

here, their true difference lies entirely in how much money they spend, not how often they visit.



Age is more or less evenly distributed across the entire customer base. The segments are purely behavioral, meaning that they are driven by how the customers buy and how much they spend, rather than demographic.

Normalized Customer Segment Profiles



The radar chart highlights clear behavioral differences among the three customer segments. Cluster 0 exhibits the strongest customer value, characterized by high purchase amounts, numerous previous purchases, and positive review ratings. Cluster 1 consists of regular but

lower-spending customers. Cluster 2 is distinguished by its significantly longer purchase intervals and lower purchase activity, suggesting these customers may require targeted retention strategies to improve engagement.

13. Recommendations

High-Value Loyal Buyers

- Offer VIP treatment through exclusive offers
- Provide exclusive products and early access to new collections
- Focus on customer retention through personalized communication
- Provide loyalty rewards

Budget-Conscious Regular Customers

- Encourage higher basket values through cross-selling and bundle offers
- Send personalized promotions
- Recommend complementary products

Occasional Customers

- Run re-engagement email campaigns
- Offer seasonal discounts
- Send reminders and personalized recommendations
- Encourage repeat purchases through exclusive offers
- Introduce first-time buyer incentives

Cluster	Profile	Priority
0	High-Value Loyal Customers	Retain and reward
1	Budget-Conscious Regular Customers	Increase spending through upselling and cross-selling
2	Occassional Customers	Reactivate with targeted marketing campaigns

14. Business Impact

This enables businesses to target specific audiences, personalize marketing campaigns, improve customer retention, and increase revenue.

